

Fire & Smoke Detection

Final results report: every model trained, every measured number, and what each experiment showed
30 September 2026

Final model (Model 1): DINOv3 ViT-S/16 backbone + Faster R-CNN detector + image-level scene classifier, trained in two stages on D-Fire.

Headline number (test split, 4,306 images)	Value
Box detection, mAP@0.5 (smoke / fire)	0.7261 (0.815 / 0.638)
Fire/smoke images detected at 0.70% false alarms (conf 0.90)	87.5%
Fire/smoke images detected at 1.30% false alarms (conf 0.85)	92.3%
False alarms on real industrial footage (54.5 min, 3 cameras, conf ≥ 0.75)	0
Best backbone, frozen-feature comparison	DINOv3 (ahead of MobileNetV3 and ResNet-18)
Best light model	MobileNetV3-L: -2.3 mAP, about 2.5x faster on CPU

What was tried and what it showed

- **Two-stage DINOv3 training** (frozen backbone, then last 2 blocks unfrozen) gave the best model: val 0.7312 -> 0.7406, test 0.7196 -> 0.7261.
- **Backbone comparison** with everything else fixed: DINOv3 0.7196 > MobileNetV3 0.6968 > ResNet-18 0.6477 on test. The choice of DINOv3 is supported by measurement.
- **4x more fire data** (DFS + FASDD, 14,792 fire images, with 1,931 leaked duplicates removed) did **not** raise the test score: best 0.7222 vs 0.7261.
- **Box mAP is limited by label consistency**, not by the model: AP drops by half when the box overlap requirement goes from 50% to 75%, and D-Fire itself labels the same smoke column as smoke in one frame and fire in the next.

1. The system and the data

Input frames are letterboxed to 640 x 640. A DINOv3 Vision Transformer (self-supervised, 21.6M parameters) produces features at four scales (strides 8/16/32/64); a Faster R-CNN head proposes and classifies boxes as *smoke* or *fire*; a second, image-level head reads the same features plus seven hand-crafted glow statistics and outputs P(smoke), P(fire) for the whole frame, so a flame hidden behind equipment can still be flagged from the light it casts. On video, an alarm requires 6 detections within 15 frames, which filters one-frame flicker.

Split (D-Fire, fixed for every experiment)	Images	Negatives	Smoke boxes	Fire boxes
Train	13,776	6,223	7,660	9,659
Validation	3,445	1,610	1,890	2,155
Test (never used for training or selection)	4,306	2,005	2,311	2,878

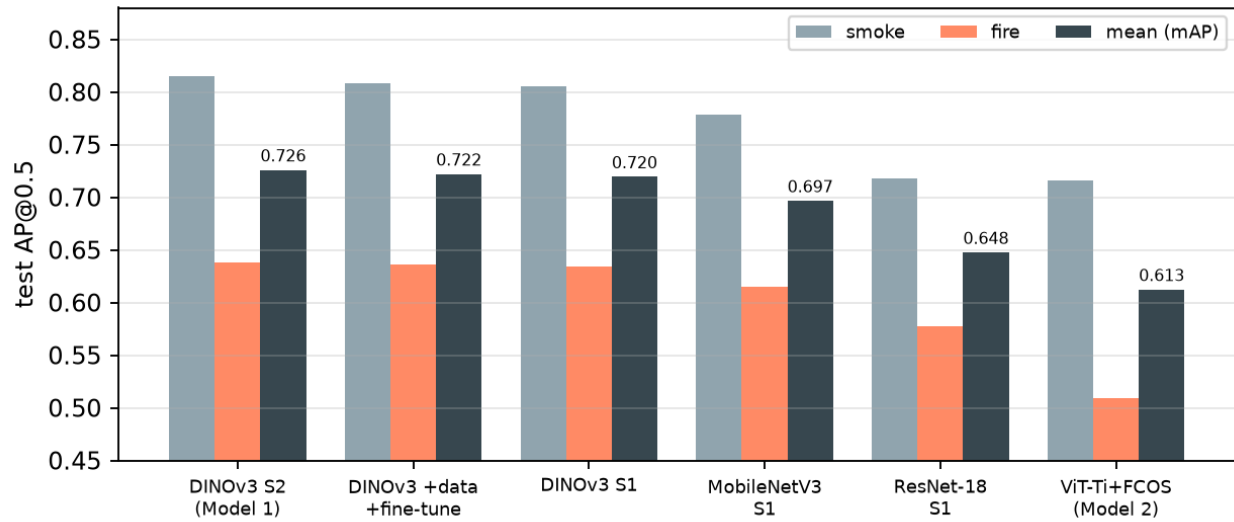
Training: AdamW, lr 1e-4 (5e-5 when unfreezing), backbone at 5% of the head learning rate, 500-step warmup then cosine decay, batch 8, mixed precision, early stopping on validation mAP@0.5; low-light, IR/grey, occlusion, crop, colour-jitter and blur augmentation. All runs on Kaggle T4 GPUs.

2. Every model trained

#	Model	Backbone state	Data	Epochs	Best val	Test mAP
1	DINOv3 ViT-S + Faster R-CNN, stage 1	frozen	D-Fire	40	0.7312	0.7196
2	DINOv3 ViT-S + Faster R-CNN, stage 2 (Model 1)	last 2 blocks	D-Fire	15	0.7406	0.7261
3	DINOv3 ViT-Ti + FCOS (Model 2, light)	frozen	D-Fire	45	0.6220	0.6125
4	MobileNetV3-L + Faster R-CNN, stage 1	frozen	D-Fire	40	0.7056	0.6968
5	ResNet-18 + Faster R-CNN, stage 1	frozen	D-Fire	40	0.6567	0.6477
6	DINOv3, stage 3 from #2	last 2 blocks	D-Fire+DFS+FASDD	12	0.7177	-
7	DINOv3, fine-tune from #6 (epoch 4)	last 2 blocks	D-Fire	4	0.7251	0.7222

Validation picks the epoch; test is reported once per model. #7's best-validation epoch (epoch 1) scored 0.7128 on test; epoch 4 is shown because it was better on both splits' alarm metrics.

3. Test-split results

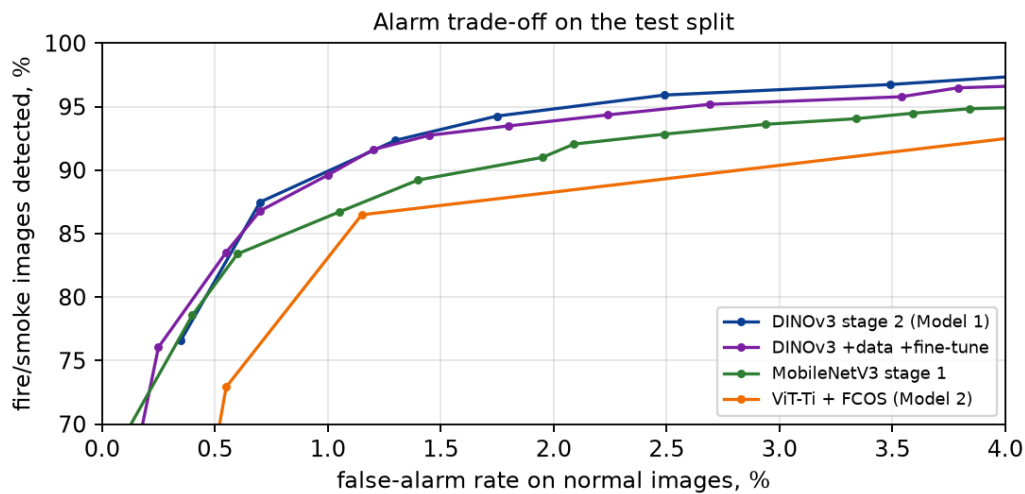


AP@0.5 per class on the 4,306-image test split. Model 1 (left) is the best on every class.

Model	Smoke AP	Fire AP	mAP@0.5	mAP@0.75	mAP@.5:.95
DINOv3 stage 2 (Model 1)	0.8146	0.6377	0.7261	0.3208	0.3644
DINOv3 + data + fine-tune	0.8082	0.6362	0.7222	0.3338	0.3665
DINOv3 stage 1	0.8050	0.6342	0.7196	0.3252	0.3627
MobileNetV3-L stage 1	0.7786	0.6150	0.6968	0.2714	0.3334
ResNet-18 stage 1	0.7177	0.5778	0.6477	0.2426	0.3046
ViT-Ti + FCOS (Model 2)	0.7158	0.5092	0.6125	0.2205	0.2854

4. Alarm behaviour: the number that decides deployment

A deployed system raises an alarm per frame, not per box. The test sweep below counts, for each confidence threshold, how many *normal* images produced any box (false-alarm rate) and how many fire/smoke images were detected.



Model (test split)	Threshold	FP images	FP rate	Detected (any)	Fire	Smoke
DINOv3 stage 2 (Model 1)	0.90	14	0.70%	87.5%	78.0%	83.9%
DINOv3 stage 2 (Model 1)	0.85	26	1.30%	92.3%	85.7%	89.9%
DINOv3 + data + fine-tune	0.90	14	0.70%	86.8%	78.8%	82.6%
DINOv3 + data + fine-tune	0.85	24	1.20%	91.6%	87.0%	88.6%
DINOv3 stage 1	0.90	14	0.70%	84.8%	75.9%	81.2%
MobileNetV3-L stage 1	0.90	12	0.60%	83.4%	73.3%	79.5%
ResNet-18 stage 1	0.90	13	0.65%	72.9%	61.4%	68.0%

Real industrial footage

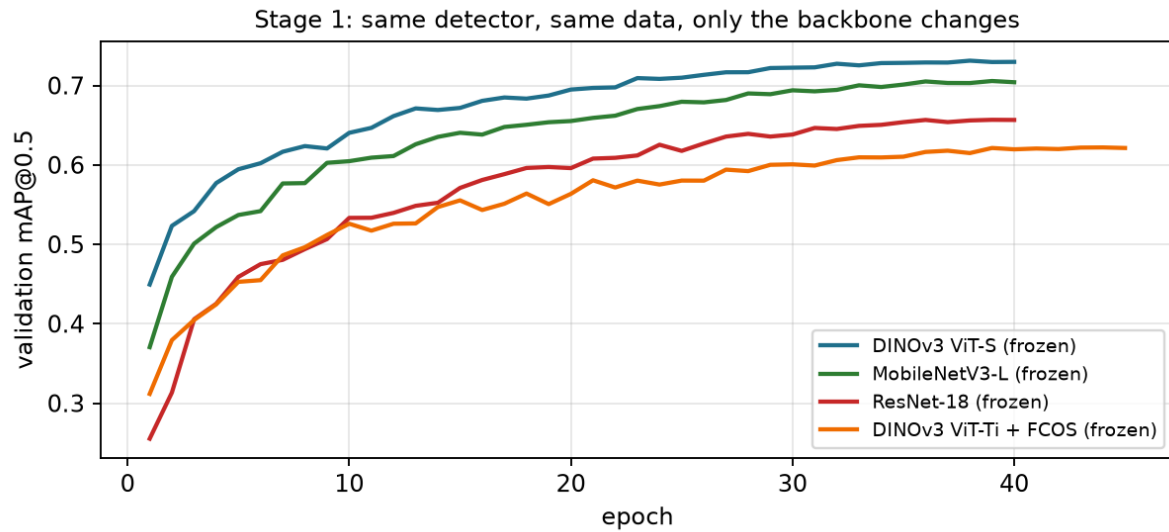
Model 1 was also run on 54.5 minutes of normal operation from three fixed industrial cameras (confidential footage, no fire present), sampled at 1 fps with the 6-of-15 confirmation rule. This measures false alarms only.

Confidence threshold	False alarms	Per hour
0.50	19	20.9
0.60	9	9.9
0.70	3	3.3
0.75	0	0.0
0.85	0	0.0
0.90	0	0.0

Zero false alarms from 0.75 upward, so the 0.85-0.90 operating point has margin. What triggered below that: bright dusty haze (as smoke) and a worker's yellow hard hat (as fire), both stationary, which is the case temporal confirmation cannot filter. They are the first candidates for hard-negative training.

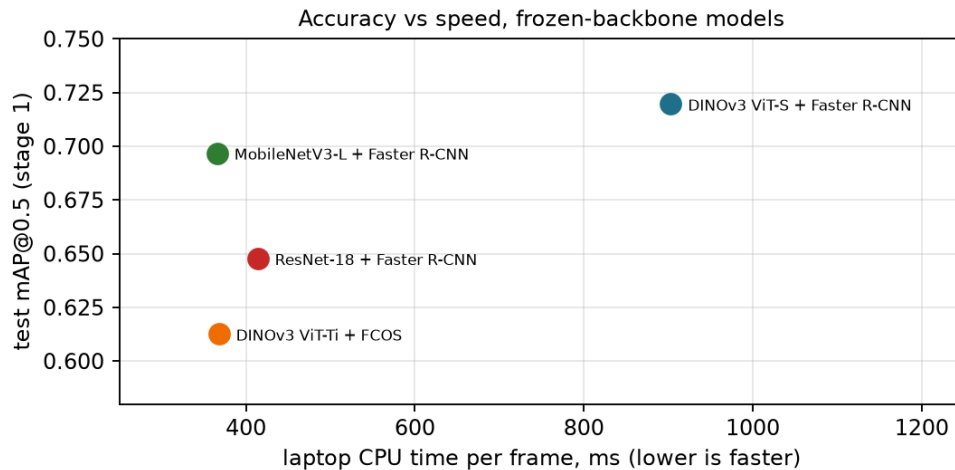
5. Backbone comparison: is DINOv3 worth it?

The same detector was trained on two ImageNet CNN backbones, each on its own code branch. Only the backbone changed: Faster R-CNN head, scene head, glow prior, augmentation, optimiser and the 40-epoch frozen-backbone schedule are identical. With the backbone frozen, the comparison tests the quality of the pretrained features directly.



Backbone (stage 1)	Params	Val mAP	Test mAP	Test fire AP	Detected @0.90	CPU ms/frame
DINOv3 ViT-S	39.3M	0.7312	0.7196	0.6342	84.8%	894-912
MobileNetV3-L	20.0M	0.7056	0.6968	0.6150	83.4%	346-385
ResNet-18	27.9M	0.6567	0.6477	0.5778	72.9%	414
DINOv3 ViT-Ti + FCOS	6.9M	0.6220	0.6125	0.5092	n/a*	338-398

CPU times: project laptop, batch 1, 640 px, fp32, median of 15 runs; ranges where two sessions differed. * Model 2's FCOS head compresses scores (max about 0.75), so 0.90 is not a usable threshold for it; at a matched 0.7% false-alarm rate it detects 76.3%.



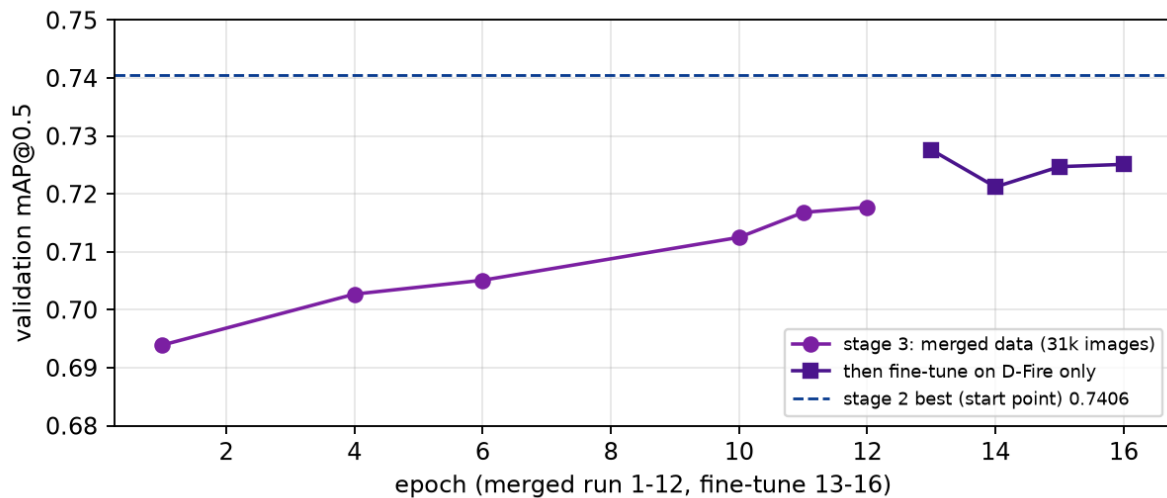
- **DINOv3 wins on accuracy** on both splits and both classes: +2.3 mAP over MobileNetV3 and +7.2 over ResNet-18 on test.
- **MobileNetV3 is the best light option**: at the speed of Model 2 it scores +8.4 mAP, and it is close to DINOv3 at the alarm threshold.
- **ResNet-18 is dominated**: slower than MobileNetV3 and clearly less accurate.

6. Experiment: 4x more fire data

Fire appeared in half as many training images as smoke (3,819 vs 6,778), and fire AP was the weakest number. Two public datasets were added to the training split only, through a merge tool that maps class names explicitly and removes near-duplicates of every existing image (perceptual hash, confirmed pixel by pixel).

	DFS	FASDD_CV
Fire images added	6,792	8,000 (capped)
Negatives added	1,008 hard negatives (lamps, sunsets...)	
Duplicates of D-Fire test images removed	34	341
Duplicates of D-Fire val / train removed	26 / 87	290 / 1,153

Training set 13,776 -> 31,076 images. Without the duplicate check, 375 test images would have entered training and inflated the test score.



Test split	Smoke AP	Fire AP	mAP@0.5	Detected @0.90 (FP images)
DINOv3 stage 2 (start point)	0.8146	0.6377	0.7261	87.5% (14)
After merged data + D-Fire fine-tune	0.8082	0.6362	0.7222	86.8% (14)

Result. The extra data did not improve the D-Fire test score; the best checkpoint ties stage 2 (0.7222 vs 0.7261, same number of false-alarm images at 0.90). Validation first dropped by 4.7 points on the mixed data - mostly smoke - and only partly recovered after a D-Fire-only fine-tune. The external datasets draw fire and smoke boxes differently, and the D-Fire benchmark rewards D-Fire's own convention. Whether the broader data helps on unfamiliar industrial scenes is not measurable on D-Fire and is the next test.

7. Why box mAP levels off around 0.72-0.74

mAP@0.5 counts a detection as correct only if its box overlaps the annotator's box by at least 50%. Tightening that to 75% roughly halves every score, which means the model usually finds the object but disagrees with the annotator about its edges - and fire and smoke have no sharp edges.

Model 1, test split	Overlap $\geq 50\%$	Overlap $\geq 75\%$
Smoke	0.815	0.405
Fire	0.638	0.236
Mean	0.726	0.321



Two D-Fire training frames taken 10 seconds apart (timestamps visible). The same thin smoke column is labelled **smoke** (blue, left) and **fire** (red, right). A model cannot score above the consistency of its labels.

- Fire is scored lower than smoke even though it is easier to see: several flames are boxed as one region by some annotators and one box per flame by others, and a correct single box then counts as one hit and several misses.
- Training has levelled off: stage 2 gained about 0.001 mAP per epoch at the end, and 4x more data did not move the test score.
- At the frame level, which is what triggers an alarm, Model 1 already detects 92.3% of fire/smoke images at a 1.3% false-alarm rate.

8. Conclusions and next steps

- **Model 1 (DINOv3 stage 2) is the final model:** best on every test metric, zero false alarms on real industrial footage at the operating thresholds.
- **DINOv3 is the right backbone:** it beat both ImageNet CNNs with everything else held fixed.
- **MobileNetV3 is the candidate for scaling to many cameras:** close to DINOv3 at the alarm threshold at about 2.5x the speed. A MobileNetV3 + FCOS version would add batched multi-camera export.
- **Next data work should target label consistency and the deployment domain,** not volume: (a) test Model 1 and the fine-tuned model on industrial video with and without fire, (b) add hard negatives from site footage (hard hats, haze), (c) label a small in-house set from the site's own cameras with one consistent box rule.
- Test-time augmentation and a DINOv3 + MobileNetV3 ensemble are available accuracy levers that need no retraining.